

CANCER DIAGNOSTICS Using MEMS And Microfluidics

With MEMS sensors at the forefront of cancer diagnostics, we stand on the cusp of a new era of precision medicine and improved patient outcomes in the fight against cancer. These tiny devices empower healthcare professionals to detect cancer cells earlier, monitor treatment response in real time, and offer personalised care



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MEMS (micro-electromechanical systems) and microfluidics are two exciting fields of technology that have converged remarkably in recent years. MEMS devices are miniature systems made of tiny mechanical and electrical components, often using semiconductor manufacturing techniques. On the other hand, microfluidics involves the manipulation and control of small amounts of fluids in microscale channels and chambers.

The fusion of MEMS and microfluidics has given rise to a new class of devices with extraordinary capabilities. These inte-

grated systems leverage the precision and scalability of MEMS technology to manipulate fluids at the micro scale, enabling applications in various domains such as biomedical diagnostics, drug delivery, chemical analysis, and lab-on-a-chip systems. By combining microfluidic channels and chambers with MEMS components like sensors, actuators, and valves, these devices can perform tasks like sample handling, mixing, separation, and analysis with unprecedented efficiency.

The integration of MEMS and microfluidics offers benefits such as reduced sample volumes, faster analysis times,

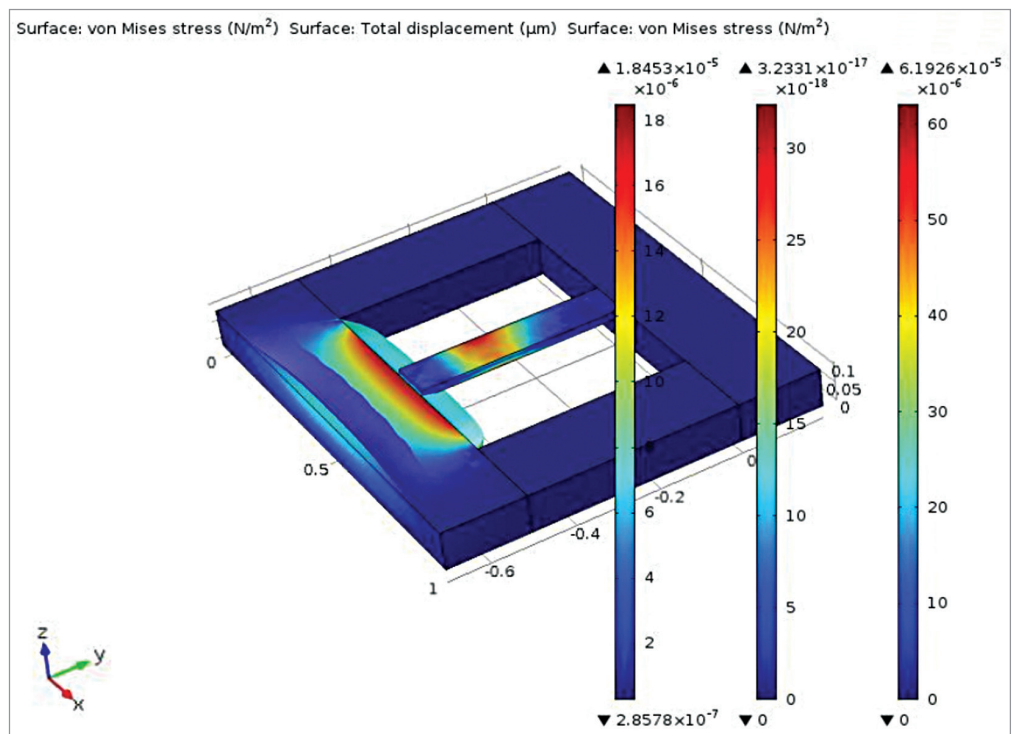


Fig. 1: Investigation on MEMS-based thermal sensor for cancer detection